

6. (a) Nuclear fusion and nuclear fission reactions both produce heat energy. Describe and compare controlled nuclear fission and nuclear fusion reactions. [6 QER]

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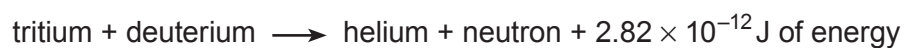
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- (b) A typical fusion reaction between tritium and deuterium is:



Calculate the number of fusion reactions needed each second to provide an energy output of 6.2 MJ every second. [2]

Number of reactions =

